



Nebraska Public Power District

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10 CFR 50.73

NLS2024044

June 6, 2024

ATTN: Document Control Desk
U.S. Nuclear Regulatory Commission
Washington, D.C. 20555-0001

Subject: Licensee Event Report No. 2023-002-01
Cooper Nuclear Station, Docket No. 50-298, DPR-46

Dear Sir or Madam:

The purpose of this correspondence is to forward Licensee Event Report 2023-002-01.

This letter does not contain regulatory commitments.

Sincerely,

Khalil Dia
Site Vice President

/jo

Attachment: Licensee Event Report 2023-002-01

cc: Regional Administrator w/attachment
USNRC - Region IV

NPG Distribution w/attachment

Cooper Project Manager w/attachment
USNRC - NRR Plant Licensing Branch IV

INPO Records Center w/attachment
via IRIS entry

Senior Resident Inspector w/attachment
USNRC - CNS

SORC Chairman w/attachment

SRAB Administrator w/attachment

CNS Records w/attachment

COOPER NUCLEAR STATION

72676 648A Ave / P.O. Box 98 / Brownville, NE 68321
<http://www.nppd.com>

NRC FORM 366 (04-02-2024)			U.S. NUCLEAR REGULATORY COMMISSION			APPROVED BY OMB: NO. 3150-0104			EXPIRES: 04/30/2027			
LICENSEE EVENT REPORT (LER) (See Page 2 for required number of digits/characters for each block) (See NUREG-1022, R.3 for instruction and guidance for completing this form http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/)												
1. Facility Name Cooper Nuclear Station						☒ 050 ☐ 052			2. Docket Number 298		3. Page 1 of 5	
4. Title Secondary Containment Differential Pressure Perturbation Exceeds Technical Specifications												
5. Event Date			6. LER Number			7. Report Date			8. Other Facilities Involved			
Month	Day	Year	Year	Sequential Number	Revision No.	Month	Day	Year				
10	22	2023	2023	002	01	06	06	2024				
9. Operating Mode 1						10. Power Level 100						
11. This Report is Submitted Pursuant to the Requirements of 10 CFR §: (Check all that apply)												
10 CFR Part 20		☐ 20.2203(a)(2)(vi)		10 CFR Part 50		☐ 50.73(a)(2)(ii)(A)		☐ 50.73(a)(2)(viii)(A)		☐ 73.1200(a)		
☐ 20.2201(b)		☐ 20.2203(a)(3)(i)		☐ 50.36(c)(1)(i)(A)		☐ 50.73(a)(2)(ii)(B)		☐ 50.73(a)(2)(viii)(B)		☐ 73.1200(b)		
☐ 20.2201(d)		☐ 20.2203(a)(3)(ii)		☐ 50.36(c)(1)(ii)(A)		☐ 50.73(a)(2)(iii)		☐ 50.73(a)(2)(ix)(A)		☐ 73.1200(c)		
☐ 20.2203(a)(1)		☐ 20.2203(a)(4)		☐ 50.36(c)(2)		☐ 50.73(a)(2)(iv)(A)		☐ 50.73(a)(2)(x)		☐ 73.1200(d)		
☐ 20.2203(a)(2)(i)		10 CFR Part 21		☐ 50.46(a)(3)(ii)		☐ 50.73(a)(2)(v)(A)		10 CFR Part 73		☐ 73.1200(e)		
☐ 20.2203(a)(2)(ii)		☐ 21.2(c)		☐ 50.69(g)		☐ 50.73(a)(2)(v)(B)		☐ 73.77(a)(1)		☐ 73.1200(f)		
☐ 20.2203(a)(2)(iii)				☐ 50.73(a)(2)(i)(A)		☒ 50.73(a)(2)(v)(C)		☐ 73.77(a)(2)(i)		☐ 73.1200(g)		
☐ 20.2203(a)(2)(iv)				☐ 50.73(a)(2)(i)(B)		☒ 50.73(a)(2)(v)(D)		☐ 73.77(a)(2)(ii)		☐ 73.1200(h)		
☐ 20.2203(a)(2)(v)				☐ 50.73(a)(2)(i)(C)		☐ 50.73(a)(2)(vii)						
☐ OTHER (Specify here, in abstract, or NRC 366A).												
12. Licensee Contact for this LER												
Licensee Contact Linda Dewhirst, Regulatory Affairs and Compliance Manager								Phone Number (include area code) (402) 825-5416				
13. Complete One Line for each Component Failure Described in this Report												
Cause	System	Component	Manufacturer	Reportable to IRIS	Cause	System	Component	Manufacturer	Reportable to IRIS			
B	NG	CDMP		Y								
14. Supplemental Report Expected ☒ No ☐ Yes (If yes, complete 15. Expected Submission Date)						15. Expected Submission Date Month: Day: Year:						
16. Abstract (Limit to 1326 spaces, i.e., approximately 13 single-spaced typewritten lines) On October 22, 2023, Reactor Building pressure exceeded the Technical Specifications limit of -0.25 inches of water gauge (wg) resulting in Secondary Containment being declared inoperable. Operations personnel walked down the ventilation systems and found that the first discharge duct hatch for EF-R-1B was open. The hatch was subsequently closed, and Secondary Containment pressure returned to the normal operating set point of -0.33 inches wg. The cause of the event was the opened first discharge duct hatch for EF-R-1B. The cause of the open hatch has been determined to be looseness of the latching mechanism likely due to normal vibrations from system operations. There was no impact on nuclear safety, plant reliability, radiological safety, or industrial safety. Event Notification 56811 was submitted on October 22, 2023, due to the unplanned inoperability of Secondary Containment.												

**LICENSEE EVENT REPORT (LER)
CONTINUATION SHEET**

(See NUREG-1022, R.3 for instruction and guidance for completing this form
<http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/>)

Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the FOIA, Library, and Information Collections Branch (T-6 A10M), U. S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by email to Infocollects.Resource@nrc.gov, and the OMB reviewer at: OMB Office of Information and Regulatory Affairs, (3150-0104), Attn: Desk Officer for the Nuclear Regulatory Commission, 725 17th Street NW, Washington, DC 20503. The NRC may not conduct or sponsor, and a person is not required to respond to, a collection of information unless the document requesting or requiring the collection displays a currently valid OMB control number.

1. FACILITY NAME	2. DOCKET NUMBER	3. LER NUMBER		
		YEAR	SEQUENTIAL NUMBER	REV NO.
		2023	002	01

NARRATIVE**PLANT STATUS**

Cooper Nuclear Station was in Mode 1, Power Operation, at 100 percent power at the time of the event on October 22, 2023.

Both Standby Gas Treatment (SGT) [EIS: BH] trains were in normal standby lineup. Reactor Building Heating and Ventilation (RB HV) [EIS: VA] was controlling Secondary Containment pressure. Single exhaust fan EF-R-1B was in service, with an indicated exhaust flow rate of 35,000 standard cubic feet per minute, for cold weather operation.

BACKGROUND

Secondary Containment serves as a barrier to confine and monitor potential releases during fuel handling operations and is a system that limits the release of radioactive materials to the environment and consists of four subsystems. One of the subsystems is the Reactor Building [EIS: NG], which encloses the Reactor Pressure Vessel [EIS: RPV] and Primary Containment [EIS: NH].

The Secondary Containment area (most of the Reactor Building) has supply and exhaust ventilating systems. The supply system furnishes filtered 100% outdoor air to all floors of the building through ductwork. The supply unit has an operating and standby (100% capacity) fan, each with vortex dampers [EIS: CDMP], which deliver supply air at a rate proportional to a static pneumatic signal set manually by operations personnel.

The exhaust air is induced from the ventilated areas to a common plenum connected to the two exhaust fans, each of 100% capacity. The air is then exhausted to the atmosphere.

During normal plant operation, a minimum average negative pressure (vacuum) of 0.25 inches of water gauge (wg) is maintained by differential pressure (DP) controllers which receive signals proportional to the pressure difference between outside air and the Secondary Containment atmosphere, and control the position of the exhaust fan vortex dampers. The DP is detected by four separate probes, with one on each side of the Reactor Building. Control action is initiated from the average value of the four sensor probes.

If a Loss of Coolant Accident should occur, all ventilation systems of the Primary Containment area and Secondary Containment area will trip, Secondary Containment is isolated automatically, and the SGT system will be automatically initiated.

Each exhaust fan has a transition piece between the round fan barrel and the following rectangular duct work. Each exhaust fan has an air operated discharge damper installed in the duct work immediately downstream of the transition piece. On each exhaust fan discharge duct, there is a duct hatch on the upstream side of the

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discharge damper, and a duct hatch on the downstream side of the discharge damper. These hatches are on the side of the duct, hinged for horizontal swing, with two securing latches. There are also access hatches on top of each of the exhaust ducts in the overhead of the refuel floor. These exhaust ducts must contain positive pressure relative to building ambient air and relative to atmosphere. If a hatch fails to stay closed, some of the exhaust air flows out of the duct and re-circulates. This causes reduced mass air flow to atmosphere, an initial increase in Secondary Containment pressure, and reduced vacuum (reduced DP to atmosphere).

EVENT DESCRIPTION

On October 22, 2023, at 1149 hours, control room operators noted that Reactor Building pressure had exceeded the Technical Specification (TS) Limiting Condition for Operation (LCO) of -0.25 inches wg. Maximum Secondary Containment pressure was -0.13 inches wg. Reactor Building Heating and Ventilation (RB HV) was configured for cold weather operation and low air flow rate, with a single exhaust fan, EF-R-1B, in service. Due to the pressure transient being above -0.25 inches wg, control room operators declared Secondary Containment inoperable at 1150 hours, and entered TS LCO 3.6.4.1, Condition A, Required Action A.1, to restore Secondary Containment to operable status. The Reactor Building Ventilation System responded without operator action to restore DP to within TS limits at 1151 hours.

Plant conditions and meteorological data were reviewed to determine the cause of the DP transient. The results of this review determined that the transient was not attributable to meteorological conditions. Operators also determined that the transient was not attributable to equipment operations, maintenance, or plant manipulation, such as RB HV supply unit air filter changes, or initiating or securing auxiliary steam to the RB HV supply unit.

Operations personnel continued to investigate by walking down the ventilation systems. At 1310 hours, operations personnel found that the first discharge duct hatch for EF-R-1B was open. The hatch was subsequently closed and Secondary Containment pressure returned to the normal operating set point of -0.33 inches wg. Operations personnel declared Secondary Containment OPERABLE and exited LCO 3.6.4.1, Condition A.

BASIS FOR REPORT

This event is reportable under 10 CFR 50.73(a)(2)(v)(C and D) – An event or condition that could have prevented the fulfillment of the safety function of structures or systems that are needed to control the release of radioactive material or mitigate the consequences of an accident. Event Notification 56811 was submitted on October 22, 2023, due to the unplanned inoperability of Secondary Containment.



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SAFETY SIGNIFICANCE

This event is non-consequential. The pressure excursion in the Reactor Building resulted in an unexpected plant condition and met the entry condition for required action pursuant to Technical Specifications. This event resulted in failure to meet Secondary Containment TS LCO 3.6.4.1 for approximately 80 seconds. Secondary Containment pressure remained sub-atmospheric throughout the event. The non-essential reactor building HVAC system automatically restored required pressure (simple restoration).

An engineering review demonstrated that the perturbation on Secondary Containment pressure caused by the non-essential HVAC system did not impact the integrity of Secondary Containment or the ability of SGT to maintain Secondary Containment at a negative pressure. The first discharge duct hatch for EF-R-1B is located inside the Secondary Containment isolation boundary and would be isolated in an accident.

As such, this event will not be counted as a Safety System Functional Failure for the Nuclear Regulatory Commission performance indicator since no loss of safety function occurred.

CAUSE

The cause of the event was the opened first discharge duct hatch for EF-R-1B. The cause of the open hatch has been determined to be looseness of the latching mechanism likely due to normal vibrations from system operations.

CORRECTIVE ACTIONS

Operations personnel closed and verified the duct hatch was secured.

Cooper Nuclear Station will examine RB HV ventilation ductwork hatches and repair as needed.

PREVIOUS EVENTS

On May 23, 2022 – Secondary Containment was declared inoperable due to a rise in differential pressure. This was reported under LER 2022-001-00 as a loss of safety function under 10 CFR 50.73(a)(2)(v)(C and D) – An event or condition that could have prevented the fulfillment of the safety function of structures or systems that are needed to control the release of radioactive material or mitigate the consequences of an accident.

On February 9, 2021 – Secondary Containment was declared inoperable due to a rise in differential pressure. This was reported under LER 2021-001-00 as a loss of safety function under 10 CFR 50.73(a)(2)(v)(C and D) – An event or condition that could have prevented the fulfillment of the safety function of structures or systems that are needed to control the release of radioactive material or mitigate the consequences of an accident.



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On November 3, 2020 – Secondary Containment was declared inoperable due to a rise in differential pressure. This was reported under LER 2020-004-00 as a loss of safety function under 10 CFR 50.73(a)(2)(v)(C and D) – An event or condition that have could have prevented the fulfillment of the safety function of structures or systems that are needed to control the release of radioactive material or mitigate the consequences of an accident.

On August 6, 2020 – Secondary Containment was declared inoperable due to a rise in differential pressure. This was reported under LER 2020-002-00 as a loss of safety function under 10 CFR 50.73(a)(2)(v)(C and D) – An event or condition that could have prevented the fulfillment of the safety function of structures or systems that are needed to control the release of radioactive material or mitigate the consequences of an accident.

On May 1, 2020 – Secondary Containment was breached due to both airlock doors in the Reactor Building being inadvertently open simultaneously. This was reported under LER 2020-001-00 as a loss of safety function under 10 CFR 50.73(a)(2)(v)(C and D) – An event or condition that could have prevented the fulfillment of the safety function of structures or systems that are needed to control the release of radioactive material or mitigate the consequences of an accident.

On August 8, 2019 – Secondary Containment was declared inoperable due to a rise in differential pressure. This was reported under LER 2019-002-00 as a loss of safety function under 10 CFR 50.73(a)(2)(v)(C and D) – An event or condition that could have prevented the fulfillment of the safety function of structures or systems that are needed to control the release of radioactive material or mitigate the consequences of an accident.

On January 6, 2014 – Secondary Containment was declared inoperable due to a rise in differential pressure. This was reported under LER 2014-001-00 as a loss of safety function under 10 CFR 50.73(a)(2)(v)(C and D) – An event or condition that could have prevented the fulfillment of the safety function of structures or systems that are needed to control the release of radioactive material or mitigate the consequences of an accident.